

Key Performance Indicators and Benchmarking

Water Utilities System Strengthening

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WATER UTILITY

KEY PERFORMANCE INDICATORS & BENCHMARKING

INDICATORS & METRICS



DATA ANALYSIS & BENCHMARKING

PERFORMANCE ANALYSIS

INDUSTRY COMPARISON

IMPROVEMENTS & GOALS

EFFICIENCY & SUSTAINABILITY

REPORTING & KPIS

TRACKING PROGRESS

BEST PRACTICES

OPTIMIZING PERFORMANCE

Strategic Value of KPIs

- **Management Tool:** Core instrument for Results-Based Management (RBM) and transparent resource allocation.
- **Rapid Assessment:** Offers a high-velocity, cost-effective diagnostic compared to exhaustive technical audits.
- **Data-Driven Planning:** Provides empirical evidence for strategic planning and capital project design.
- **Multi-Stakeholder Language:** Bridges communication between utilities, regulators, donors, and customers.

SMART Indicators for Water Utilities

- **Specific:** Clearly target distinct aspects of water operations (e.g., Non-Revenue Water (NRW) or water quality compliance) rather than vague goals.
- **Measurable:** Quantifiable, reliable, and unambiguous, allowing for objective comparison across different service areas.
- **Attainable:** Realistic and cost-effective to collect, ensuring the utility has the technical capacity and budget to track them accurately.
- **Relevant:** Align with strategic utility priorities, such as service continuity, customer satisfaction, or financial sustainability.
- **Time-bound:** Sensitive to change over specific periods (e.g., monthly, quarterly, or annually) to allow for effective trend analysis and reporting.

Strategic KPI Framework (The Three Pillars)

- **Water Service Quality:** Measuring reliability, water safety, and customer redressal efficiency.
- **Economic Efficiency & Financial Viability:** Evaluating financial viability through billed tariffs, collection efficiency, and cost recovery.
- **Operational & Infrastructure Efficiency:** Assessing long-term health via NRW (Non-Revenue Water) management and asset maintenance.

Stakeholder Benefits

- **Regulators:** Transparent basis for tariff adjustments and compliance monitoring.
- **Utility Managers:** Identifies operational "bottlenecks" requiring immediate intervention.
- **Donors/Investors:** Demonstrates institutional creditworthiness and measurable ROI.
- **Customers:** Ensures accountability and sets a clear roadmap for improved service delivery.

Performance Benchmarking

Benchmarking enables utilities to identify:

- i. What they are doing?
- ii. How they are doing it?
- iii. How others do it?
- iv. How well they are doing it with reference to standards and best practices?
- v. What and how to improve?

Concepts of Benchmarking for Utility:

- a) Better service quality
- b) Higher resources productivity
- c) Higher satisfaction by consumers
- d) Higher general return for the company
- e) Environment improvement
- f) Basically higher operational efficiency

Benchmarking Objectives

- **Institutional Evolution:** Transitioning from reactive maintenance to proactive performance growth.
- **Performance Diagnostics:** Benchmarking internal history against national peers and international best practices.
- **Internal Governance:** Establishing standardized Managerial, Economic, and Operational KPIs for leadership guidance.
- **Gap Identification:** Identifying inefficiencies to adopt more effective methodologies aligned with the corporate business plan.

Water Service Quality KPIs

Focuses on accessibility, reliability, and customer satisfaction.

No.	KPIs	Unit
KPI 1	Water Service Coverage	%
KPI 2	Water Demand Coverage	%
KPI 3	Water Supply Continuity	Hours/Day
KPI 4	Residential Per Capita Consumption	Liters/Capita/Day
KPI 5	Increase in Domestic Connection	%
KPI 6	Quality of Water Supplied Compliance	%
KPI 7	Efficiency in Redressal of Customer Complaints	%

Operational & Infrastructure Efficiency KPIs

Focuses on asset health, resource management, and technical efficiency.

No.	KPIs	Unit
KPI 8	Non-Revenue Water	%
KPI 9	Distribution Non-Revenue Water	m3/km/day
KPI 10	Non-Revenue Water per Connection	m3/conn/d
KPI 11	Mains Failure Rate (Bursts per 100km)	%
KPI 12	Estimated Yield against Water Demand	%
KPI 13	Estimated Yield against Installed Capacity	%
KPI 14	Actual Water Production against Installed Capacity	%
KPI 15	Personnel Intensity (Staffing Efficiency)	No.
KPI 16	Percentage of Functional Water Meters	%
KPI 17	Meters Checked, Recalibrated, and Replaced	%

Economic Efficiency & Financial Viability KPIs

Focuses on financial viability, cost recovery, and fiscal management.

No.	KPIs	Unit
KPI 18	Operating Cost Coverage Ratio (OCCR)	%
KPI 19	Average Billed Tariff	Birr/m ³
KPI 20	Revenue Collection Efficiency	%
KPI 21	Commercial Efficiency Index (CEI)	%
KPI 22	Personnel Expenditure as Percentage of Revenue Collection	%
KPI 23	Working Ratio (Tariff Cost Recovery)	%
KPI 24	Water Production Cost per Cubic Meter	Birr/m ³
KPI 25	Energy Costs as a Percent of Total O and M Costs	%
KPI 26	Maintenance Costs as a Percent of Total O and M Costs	%
KPI 27	Debt Service Ratio (DSR)	%

Water Service Quality KPIs

Focuses on accessibility, reliability, and customer satisfaction.

Pillar	Focus Area	Associated KPIs
Access & Growth	Expanding the network footprint.	Water Service Coverage; Increase in Domestic Connections.
Operational Reliability	Meeting demand and maintaining flow.	Water Demand Coverage; Water Supply Continuity; Residential Per Capita Consumption.
Quality & Service	Safety and customer satisfaction.	Quality of Water Supplied Compliance; Efficiency in Redressal of Customer Complaints.

WATER SERVICE COVERAGE



Population Served
Coverage Area

WATER DEMAND COVERAGE



Supply vs Demand
Meeting Water Needs

WATER SUPPLY CONTINUITY



Hours of Supply
Uninterrupted Service

RESIDENTIAL PER CAPITA CONSUMPTION



Water Usage
Per Person

INCREASE IN DOMESTIC CONNECTIONS



New Household Connections
Expanding Network

WATER QUALITY COMPLIANCE



Water Testing
Safety Standards

KPI 1: Water Service Coverage (WSC)

- Proportion of the population within a utility's designated service area that has access to an improved water source.
- Integrates both direct household connections and communal access points.
- **The Master Formula:**

$$WSC = (P_{\text{served}} \div P_{\text{total}}) * 100$$

- **Where P_{served} is calculated as:**

$$P_{\text{served}} = (B * E) + (C * D * E)$$

P_{total} : Total population within the utility's legal jurisdiction (CSA data).

B: Total number of active domestic metered connections.

C: Total number of functional public water points.

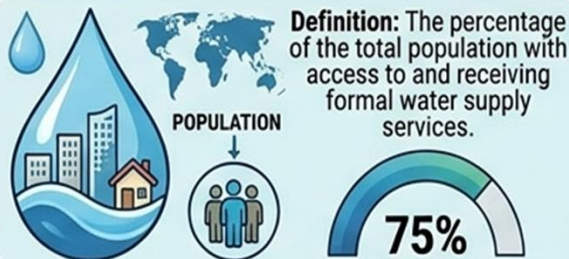
D: Estimated number of households served per public water point.

E: Average Household Size (Average family size per CSA).

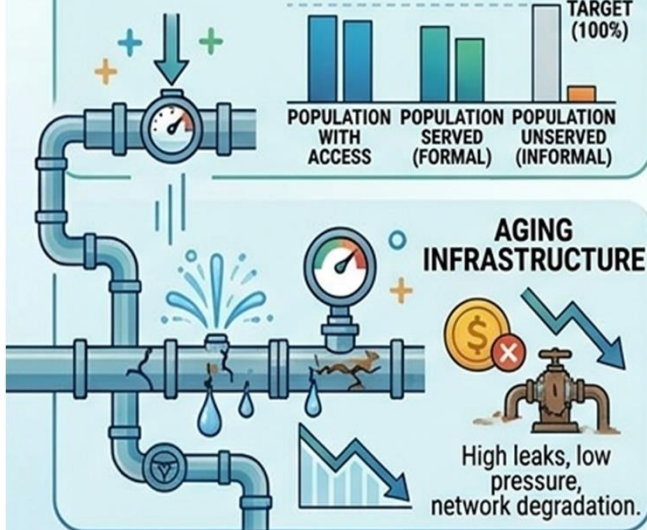
KPI 1: Water Service Coverage (WSC)

WATER SERVICE COVERAGE (WSC) COMPONENTS & IMPROVEMENT

WSC OVERVIEW & DEFINITION



CHALLENGES TO EXPANDING WSC

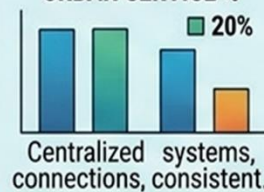


SERVICE TYPOLOGIES & INFRASTRUCTURE

URBAN NETWORKED SUPPLY



URBAN SERVICE %



RAPID URBANIZATION

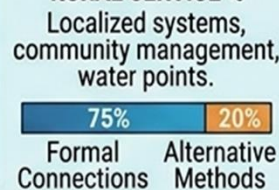
Unplanned growth, high demand pressure, informal settlements.



RURAL & DECENTRALIZED SUPPLY



RURAL SERVICE %



FINANCIAL & OPERATIONAL CONSTRAINTS

Inefficient operations.



IMPROVEMENT STRATEGIES

NETWORK REHABILITATION



Minimize losses, replace old assets.

DEMAND MANAGEMENT



Efficient use, tiered tariffs, smart metering.

IMPROVEMENT STRATEGIES

NETWORK REHABILITATION



Minimize losses, replace old assets.

DEMAND MANAGEMENT



Efficient use, tiered tariffs, smart metering.

ACCESS EXPANSION PROJECTS



Extend network to underserved areas.

SUSTAINABLE FINANCING



Mobilize funding, private cost-effective pricing.

OVERALL GOAL: 100% WSC & RELIABILITY

KPI 1: Water Service Coverage (WSC)

- **Growth Planning:** Identifies "water deserts", allowing to prioritize CAPEX for network expansion.
- **Operational Validation:** New connections are keeping pace with population growth.
- **Revenue Forecasting:** Increasing coverage usually correlates directly with increasing the utility's revenue.
- **Regulatory Compliance:** Use this metric to determine if a utility should maintain its license or receive subsidies.
- **Health Correlation:** High WSC is the strongest predictor for reduced water-borne diseases.

KPI 1: Water Service Coverage (WSC)

Performance Level	WSC Range	Context/Implication
World Class	98% – 100%	Standard in most developed urban centers; focus is on maintenance rather than expansion.
High Performing	85% – 97%	Strong coverage; usually indicates a utility that is successfully managing rapid urbanization.
Developing	60% – 84%	Significant reliance on public water points; indicates a need for heavy infrastructure investment.
Critical	< 60%	Massive service gaps; high risk of informal, unregulated water markets and public health crises.

The IBNET typically looks for a trend of 1–2% growth annually in developing regions.

KPI 2: Water Demand Coverage

- Actual residential water consumption compares to an established regional or national efficiency target.
- Contrasts **actual consumption** (measured) with a **policy target**

Component	Description	Example Data
Actual Per Capita Consumption	Total residential water delivered divided by the population served.	500,000 L / 10,000 people = 50 L/person/day
Target Per Capita Consumption	The official daily efficiency target for your region.	80 L/person/day (as per Bahir Dar city's standard)
Water Demand Coverage	(Actual Per Capita Consumption / Target Per Capita Consumption) * 100	$(50 / 80) * 100 = 62.5\%$

KPI 2: Water Demand Coverage

- **Gap Analysis:** What people need and what they get. A widening gap signals that population growth is outpacing water production.
- **Revenue Impact:** While it measures volume delivered, it also suggests potential revenue.
- **Operational Stress:** Consistently low coverage often indicates high NRW.

Level of Coverage	General Standard/Target
Ideal	100% (Meeting or exceeding country per capita targets).
Acceptable	>95% (Allows for minor fluctuations/maintenance).
Critical	<80% (Indicates severe rationing, intermittent supply, or failing infrastructure).

KPI 3: Reliability of Supply (Supply Continuity)

- Measures the average duration (in hours per day) that a consumer has water available at their tap.

Average Continuity = Sum of (Hours of Service per Zone * Total Connections in that Zone) ÷ Total Connections in all Zones

Zone	Service Hours (Hrs/Day)	Number of Connections	Calculation (Hrs × Conn)
Zone A (Central)	18 hours	2,000	18*2,000 = 36,000
Zone B (Periphery)	4 hours	3,000	4*3,000 = 12,000
Total		5,000	48,000

$48,000 \div 5,000 = 9.6$ Hours/Day. The average customer receives 9.6 hours of water daily.

KPI 3: Reliability of Supply (Supply Continuity)

- Temporal availability of water services within a 24-hour cycle.
- The objective is to quantify the extent of intermittent versus continuous supply.

Reliability (%) = (Average Daily Supply Duration (Hours) ÷ 24 Hours)*100

- **Average Daily Supply Duration:** The mean number of hours per day during which the network is pressurized and water is available at the consumer's tap.

Reliability (%)	Service Level	Operational Context
100%	Continuous	Standard for modern urban utilities; minimizes contamination risk from back-siphonage
75% – 99%	High Intermittency	May indicate scheduled rationing or localized hydraulic bottlenecks
Below 50%	Critical Intermittency	Significant infrastructure or resource deficit; high risk to public health and pipe integrity

KPI 4: Residential Per Capita Consumption (RPC)

- The average volume of water used daily by a single resident within the utility's service area.
- Typically expressed in **Liters Per Capita per Day (LPCD)**.

$$\text{RPC} = \text{DRBV} \div \text{P}$$

- DRBV: Daily total Residential Billed Volume (Total metered consumption for residential accounts)
- P: Total Resident Population

KPI 4: Residential Per Capita Consumption (RPC)

Demand Forecasting: If the population is growing and the RPC is stable, you can project exactly when you will need to build your next water supply.

Conservation Strategy: If the RPC is high compared to similar cities, management knows to implement "Demand Side Management" rather than just building more supply.

Region/Context	Typical Range (LPCD)	Description
WHO Minimum	20 – 50	Minimum for basic hygiene and survival.
Efficient Urban	100 – 150	Common in parts of Europe and water-stressed cities.
High Consumption	250 – 400+	Common in North America or areas with heavy irrigation.

The "Ideal" Standard: There is no universal "best" number, but many sustainable cities aim for approximately 110–130 LPCD to balance comfort with environmental stewardship.

KPI 5: Increase in Domestic Connection (%)

SUMMARY METRIC: INCREASE IN DOMESTIC CONNECTION (%)

INCREASE METRIC OVERVIEW



25.0%

Calculation: Percentage growth in formal water connections for domestic properties over a specific period.

KEY GROWTH DRIVERS & CONTEXT



Drivers: Network expansion projects and new urban developments.

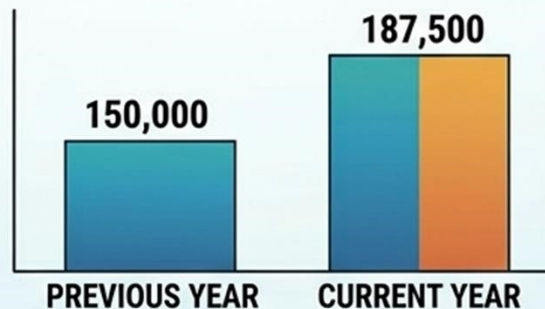


Policy: Subsidized connections, streamlined processes, and favorable tariffs.



Impact: More households with reliable access and improved health.

DOMESTIC CONNECTIONS PROGRESS



METRIC FOCUS: YEAR-ON-YEAR GROWTH IN DOMESTIC CONNECTIONS

KPI 5: Increase in Domestic Connection (%)

- Measure of system expansion, indicating whether the number of households with access to the water network is growing, stable, or declining.
- Tracks the annual rate at which new households are connected to the municipal water supply.

$$\text{(Number of New Domestic Connections in a Year} \div \text{Total Domestic Connections at Start of Year)} \times 100$$

- If a utility starts the year with 50,000 connections and adds 1,500 new domestic connections during the year, the increase is: $(1,500 / 50,000) \times 100 = 3.0\%$

KPI 5: Increase in Domestic Connection (%)

Network Planning: A high growth rate signals a need for infrastructure investment to maintain pressure and service quality.

Revenue Forecasting: Domestic connections are usually the most stable source of revenue.

Social Responsibility: Regulatory requirement to ensure "Water for All," moving populations away from unregulated or unsafe water sources.

Growth Rate	Status	Management Action
< 1%	Stagnant	A mature market or a utility struggling with "bottleneck" in infrastructure.
2% – 5%	Healthy	Standard for growing urban utilities; manageable and sustainable.
> 7%	Aggressive	Common in developing nations; requires careful monitoring to prevent system-wide pressure drops.

Always compare this KPI against the Population Growth Rate. If your connection growth is lower than the population growth, your "service coverage" is actually declining.

KPI 6: Quality of Water Supplied Compliance

- The percentage of water samples that meet or exceed national or WHO drinking water standards for physical, chemical, and bacteriological parameters.

$$C = (A \div B) * 100$$

- **A = Total number of water quality samples passed the test per year:** Samples that met all required standards (e.g., Turbidity, Residual Chlorine, E. coli).
- **B = Total number of water quality samples tested per year:** The sum of all tests conducted at the source, treatment plant, and in the distribution network.
- **C = Quality of Water Supplied Compliance (%)**

KPI 6: Quality of Water Supplied Compliance

Category	Typical Parameters	Why it matters
Microbiological	E. coli, Total Coliforms	Ensures no fecal contamination or pathogens.
Physical	Turbidity, Color, pH	Affects the appearance and the efficiency of disinfection.
Chemical	Residual Chlorine, Fluoride, Nitrates	Ensures the water remains disinfected until it reaches the tap.

- **Public Trust:** Consistently high scores prevent "boil water" advisories and build consumer confidence.
- **Regulatory Requirement:** Most national water authorities require this data to be reported to maintain operating licenses.
- A score of **less than 95%** is considered a major failure requiring immediate corrective action.

KPI 7: Efficiency in Redressal of Customer Complaints

- Complaints that are successfully resolved within a specific period relative to the total number of complaints received.

$$\text{Efficiency in Redressal} = (B \div A) * 100$$

—**A:** Total number of complaints received (the "Inflow").

—**B:** Total number of complaints redressed/resolved (the "Outflow").

- **Technical Resolution:** The physical leak is repaired, or the water quality issue is fixed.
- **Admin. Resolution:** The customer's account is updated or the billing error is corrected.
- **Communication:** The customer has been notified that the issue is closed.

KPI 7: Efficiency in Redressal of Customer Complaints

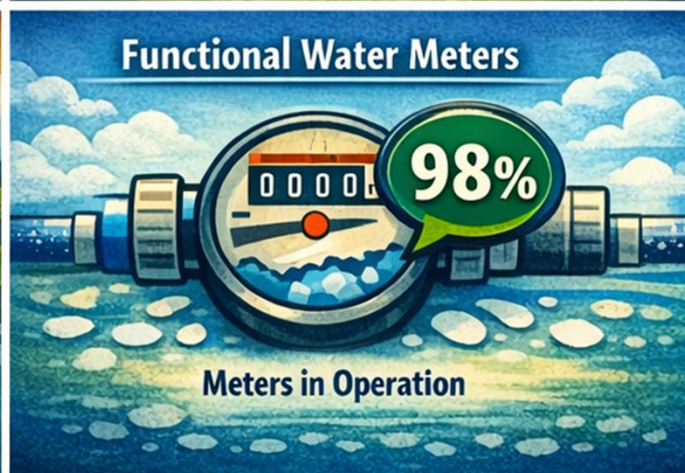
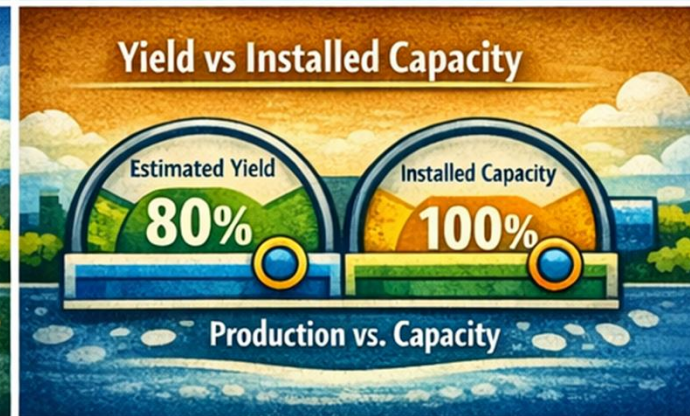
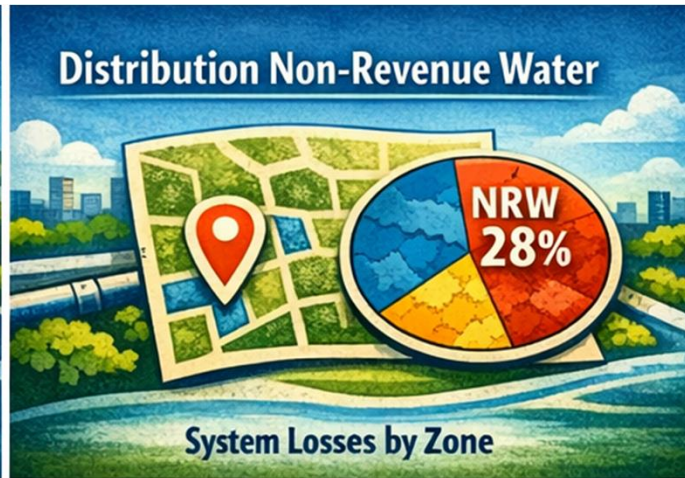
Performance Level	Efficiency Range
Excellent	> 95%
Good	80% - 95%
Needs Improvement	60% - 79%
Critical	< 60%

- **Reduces NRW:** Faster redressal of leaks prevents physical water loss.
- **Public Health:** Response to sanitation complaints prevents environmental contamination.
- **Financial Health:** Resolving billing complaints ensures stable cash flow and fewer pending payments.

Operational & Infrastructure Efficiency KPIs

Focuses on asset health, resource management, and technical efficiency.

Category	KPI Focus	Objective
Water Loss (NRW)	Distribution NRW; NRW per Connection.	To minimize physical leaks and commercial losses (theft/billing errors).
Asset Integrity	Mains Failure Rate; Functional Water Meters.	To track the physical condition of the pipes and the accuracy of billing tools.
Capacity Utilization	Yield vs. Demand; Actual Production vs. Capacity.	To ensure the water sources are neither underutilized nor dangerously overstretched.
Resource Efficiency	Personnel Intensity; Meter Maintenance.	To optimize staffing levels and ensure proactive equipment recalibration.



KPI 8: Non-Revenue Water (NRW)

- The difference between the volume of water put into a transmission and distribution system and the volume of water billed to customers.
- "Volume of NRW" must be broken down into three distinct categories:
 1. **Unbilled Authorized Consumption:** Used for legitimate purposes that is not billed (e.g., firefighting, mains flushing).
 2. **Apparent Losses (Commercial):** Consumed but not measured due to meter inaccuracies, data handling errors, or unauthorized consumption (theft).
 3. **Real Losses (Physical):** Physical lost from the system through leaks in mains, service connections, and tank overflows.

KPI 8: Non-Revenue Water (NRW%)

- NRW is expressed as a percentage of the total System Input Volume (SIV):
- $\text{NRW (\%)} = ((\text{SIV} - \text{Billed Authorized Consumption}) \div \text{SIV}) * 100$

NRW Range	Efficiency Rating	Typical Industry Status
< 10%	Excellent	World-class performance; typical of highly modernized, leak-detected networks.
10% – 25%	Acceptable	Standard for well-managed utilities in developed urban areas.
> 30%	Poor	Indicates significant infrastructure aging or systemic commercial losses (theft/meter errors).

KPI 8: Non-Revenue Water (NRW%)

NRW Management Significance

- **Resource Conservation:** Reducing NRW is often more cost-effective than developing new water sources.
- **Energy Efficiency:** Every cubic meter of lost water represents wasted electricity used for pumping and chemicals used for treatment.
- **Revenue Protection:** Identifying "Apparent Losses" (theft or meter under-registration) provides an immediate boost of revenue without increasing tariffs.

KPI 9: Distribution Non-Revenue Water (m³/km/day)

- Measures the volume of water lost daily per kilometer of the distribution network.
- A utility with a very long pipe network isn't unfairly compared to a compact city utility.

$$\text{Distribution NRW} = (\text{NRW m}^3/\text{day}) \div L$$

- **NRW:** The total volume of NRW (Real Losses + Apparent Losses + Unbilled Authorized Consumption).
- **Network Length (L):** The total linear distance of all mains and distribution network in kilometers (excluding service connections).

KPI 9: Distribution Non-Revenue Water (m³/km/day)

Allows for "Apples-to-Apples" comparisons between utilities of different sizes.

Value	Performance Tier	Engineering Implication
< 5	High Efficiency	Indicates a proactive leak detection program and high-quality pipe materials.
5 – 15	Moderate	Typical for established urban networks; requires routine pressure management.
> 20	Critical	Suggests systemic pipe failures or significant illegal tapping; infrastructure replacement is likely necessary.

- **Prioritizing Repairs:** Identify specific DMAs, ensuring maintenance budgets.
- **Investment Planning:** Strongest evidence to secure funding for replacement programs.
- **Pressure Management:** Helps monitor the success of **PRVs** in extending the life of assets.

KPI 10: Non-Revenue Water per Connection

- Quantifies the average daily NRW associated with each individual service point.
- Evaluate the performance of the most vulnerable part of the network—the service connection and the customer meter.
- Expressed in cubic meters per connection per day ($\text{m}^3/\text{conn}/\text{d}$).

$$\text{NRW per Connection} = \text{NRW in m}^3/\text{day} \div C$$

- C: Total number of active service connections (not total population).

KPI 10: Non-Revenue Water per Connection

Value (m3/conn/d)	Efficiency Tier	Primary Management Focus
< 0.1	High Efficiency	Focus on maintaining meter accuracy and rapid response to reported leaks.
0.1 – 0.3	Standard	Typical for modern urban systems. Requires routine active leak detection.
> 0.5	Low Efficiency	Likely indicates aging service pipes, widespread meter under-registration, or unauthorized usage.

- **Commercial Loss Detection:** "Apparent Losses" occur at the connection level, a high value here often points to a need for a Meter Replacement Program or improved audits.
- **Service Pipe Integrity:** More water is lost through small leaks in service pipes than through the large water mains themselves.

NON-REVENUE WATER (NRW) COMPONENTS & MANAGEMENT

OVERVIEW & DEFINITION



REAL (PHYSICAL) LOSSES

NETWORK LEAKS

Burst pipes & background leaks.

\$4.8500 / m³



OKTWICK LEAKS



NRW

Water produced that is lost before it reaches the customer and provides no revenue.



PRODUCED WATER

REVENUE WATER (Sold)

NON-REVENUE WATER (Lost)



REAL OVERFLOWS

Cost to treat and distribute 1 m³ of water.

35% / m³



SERVICE CONNECTION LEAKS

Leaks in utility and customer service lines.



TANK OVERFLOWS

Inefficient operations & controls.



APPARENT (COMMERCIAL) LOSSES

METERING INACCURACIES



UNAUTHORIZED CONSUMPTION



REDUCTION STRATEGIES

ACTIVE LEAK DETECTION



Proactive location & repair.

PRESSURE MANAGEMENT



Optimal pressure to minimize bursts.

METER REPLACEMENT & SMART METERS



Improve accuracy & data handling.

NETWORK UPGRADES



Replace aging infrastructure.

OVERALL REDUCTION GOAL: LOWER NRW %

KPI 11: Mains Failure Rate

(Bursts/100km/year)

- Quantifies the frequency of structural failures in the water distribution network, normalized by distance.
- The rate is typically calculated annually and expressed as failures per 100 kilometers of pipe.

$$\text{Mains Failure Rate} = ((\text{Total Number of Annual Mains Failures} \div \text{Total Network Length (km)}) * 100$$

- **Mains Failure:** Any structural breach of a water main (excluding service connections) that results in a loss of water or pressure and requires a repair.
- **Total Network Length:** The total linear distance of all trunk mains and distribution pipes in the system, usually verified via **GIS**

KPI 11: Mains Failure Rate (Bursts/100km/year)

Failures per 100km	Performance Tier	Asset Management Implication
< 5	Excellent	Indicates a young or well-maintained network with effective pressure control.
5 – 15	Good/Stable	Standard for mature urban networks; requires routine "find-and-fix" maintenance.
15 – 30	Degrading	High failure rates suggest the end of asset useful life; indicates a need for pipe replacement.
> 30	Critical	Significant risk to service continuity and water quality; requires urgent capital intervention.

- **Capital Investment Planning:** Provides the data necessary to transition from "Reactive Maintenance" (fixing bursts) to "Proactive Replacement" (replacing pipes before they fail).
- **Service Level Agreements:** High failure rates lead to frequent unplanned outages, negatively impacting customer satisfaction and regulatory compliance.

KPI 12: Estimated Yield against Water Demand

- Measures the sufficiency of existing water sources compared to the projected needs of the town.

$$\text{Yield against Demand} = (A \div B) * 100$$

- **A = Estimated Yield of Water Sources (m³/day):** The total reliable daily volume that can be extracted from all current sources without depleting them (safe yield).
 - **B = Average Water Demand (m³/day):** The projected daily volume needed by the town 10 years from now, accounting for population growth, industrial use, and expected losses.
- **Ratio ≥ 100%:** The utility has secured enough water to meet the town's needs for the next decade.
 - **Ratio < 100%:** There is a "Water Gap." The utility must urgently invest in new source development to avoid future water rationing or shortages.

KPI 12: Estimated Yield against Water Demand

Variable	What to Consider
Yield (A)	Use the Safe Yield (the amount you can pump sustainably even during the dry season), not just the maximum pump capacity.
Demand (B)	This should include NRW. If the town needs 8,000 m ³ for drinking, but you lose 2,000 m ³ in leaks, your demand is actually 10,000 m ³ .

- **Urban Planning:** It prevents the town from expanding residential areas into zones where the water source cannot reach.
- **Investment Justification:** If KPI is 60%, the utility has a powerful mathematical argument to ask the government or donors for funding to build new infrastructure.
- **Climate Resilience:** By monitoring the yield (A) over time, utilities can see if climate change or falling water tables are shrinking their available supply.

KPI 13: Estimated Yield against Installed Capacity

- Compares the physical capability of pumping and installed capacity against the total sustainable yield volume the environment can provide.

$$F = (D \div E) * 100$$

- **D = Installed Capacity (m³/day):** The maximum volume of water your current pumps, motors, and treatment plants are designed to produce in 24 hours.
- **E = Estimated Yield (m³/day):** The total "Safe Yield" of the water source (boreholes, rivers, etc.) as determined by hydrogeological or hydrological studies.
- **F = % Age of Installed Capacity against Yield:** The Source Utilization Efficiency.

KPI 13: Estimated Yield against Installed Capacity

- **Low Percentage (< 80%):** Source is capable of providing more water than your current pumps can handle. This is good for future expansion.
- **High Percentage (Approaching 100%):** Extracting almost everything the source can safely provide. Any further increase in pump size could damage the source
- **Over 100%: Danger Zone.** Pumps are designed to extract more water than the source can sustainably give, which leads to equipment failure or environmental depletion.

KPI 14: Actual Water Production against Installed Capacity

- Compares the volume of water *actually* produced and sent into the network against the maximum volume pumps were *designed* to produce.

$$I = (G \div H) * 100$$

- **G = Actual Water Production (m³/day):** The real volume of water measured by bulk meters leaving the treatment plant or source daily.
- **H = Installed Capacity (m³/day):** The "Nameplate" capacity. What the manufacturer says the pumps and filters can handle in 24 hours of continuous operation.
- **I = % Age of Actual Production against Installed Capacity:** The utilization rate.

KPI 14: Actual Water Production against Installed Capacity

- Measures the operational efficiency of water production facilities. How much of "machine's" theoretical potential is actually being used to deliver water.
- **High Percentage (85%–95%):** Running at near-peak efficiency. However, it also means there is very little room for maintenance without cutting off the water supply.
- **Low Percentage (< 60%):** "Underutilization." Expensive infrastructure sitting idle. This could be due to frequent power outages, broken pumps.
- **Over 100%:** "over-driving" equipment beyond its design life, which usually leads to frequent mechanical breakdowns.

KPI 15: Personnel Intensity (Staff per 1,000 Connections)

- Measures the human resource requirement relative to the utility's customer base.
- Evaluate labor productivity and determine if the utility is overstaffed or under-resourced.

Personnel Intensity = (Total Full-Time Equivalent (FTE) Personnel ÷ Total Number of Connected Properties) * 1,000

Performance Tier	Staff per 1,000 Connections	Context
High Performer	< 1.0 to 2.5	High automation; likely in developed urban centers (e.g., Singapore, Netherlands).
Standard Performer	2.5 to 5.0	Typical for well-run utilities in middle-income or developing regions.
Low Performer	> 5.0 to 10.0+	Heavy manual labor; often indicates "social employment" or lack of basic technology.

KPI 15: Personnel Intensity (Staff per 1,000 Connections)

- **Cost Control:** A high ratio usually suggests that manual processes (guards, meter reading or manual valve turning) are inflating the payroll.
- **Technological Maturity:** Apply automation; SCADA systems, GIS, and AMI (Smart Meters), this number should naturally trend downward.
- **IBNET suggest:** Many developing utilities, reaching a target of less than 5.0 is the first major milestone toward financial sustainability.
- **Recommendations:** Increase the number of connections, transition to smart infrastructure, outsourcing, training & multitasking

KPI 16: Percentage of Functional Water Meters

PERCENTAGE OF FUNCTIONAL WATER METERS

FUNCTIONALITY METRIC OVERVIEW



88.0%

Calculation: Percentage of total installed water meters that are functional and accurate.



DRIVERS & IMPACT

MAINTENANCE & CALIBRATION



Drivers: Regular meter testing, proactive maintenance, and prompt repairs.

TECHNOLOGY ADOPTION



Drivers: Replacement of aging meters with modern, remote-reading smart meters.

IMPACT: REVENUE & OPERATIONAL EFFICIENCY



Impact: Accurate billing, reduced Non-Revenue Water, and optimized distribution network.

DOMESTIC CONNECTIONS PROGRESS

FUNCTIONAL
METERS



VS



FUNCTIONAL METERS

TOTAL METERS

METRIC FOCUS: RATIO OF OPERATIONAL METERS TO TOTAL INSTALLED

KPI 16: Percentage of Functional Water Meters

Component	Description	Role in Calculation
A. Active Functional Meters	Meters that are functioning correctly and generating customer usage data for billing.	Numerator (Functional Meters)
B. Stopped/Malfunctioning Meters	Meters that is broken, inaccurate, or not registering flow. Common faults include not registering use, inaccurate readings, physical damage.	Part of the Denominator (Total Inventory)
Total Active Billable Meters	This should logically represent the total meter inventory.	Denominator (Total Inventory = A + B)
Functional Meter Percentage	The final KPI.	Result

$$\text{Functional Water Meter \%} = (A / (A + B)) \times 100$$

KPI 16: Percentage of Functional Water Meters

Performance Level	Percentage Range	Management Context
World Class	> 98%	Common in utilities with automated reading (AMR/AMI) and strict 5-10 year replacement cycles.
Acceptable	90% – 95%	Standard for well-managed utilities; indicates a functional but perhaps aging meter fleet.
At Risk	80% – 89%	Significant revenue leakage; requires immediate investment in a meter replacement program.
Critical	< 80%	Utility is likely in a "death spiral" where lack of revenue prevents fixing the meters that would generate revenue.

Reduction of NRW: Inaccurate meters lead to "Apparent Losses," where water is consumed but not billed. High functionality ensures that "Real Losses" (leaks) can be notable than "Paper Losses."

Asset Management Strategy: Dictates the meter replacement cycle. Analyzing failure rates helps management decide between reactive repairs or proactive.

KPI 17: Meters Checked, Recalibrated, and Replaced

$$\text{Percentage} = (A / B) \times 100$$

- ✓ **A** = Total number of meters checked, recalibrated, and replaced by the end of the year.
- ✓ **B** = Total number of registered customer meters in the town.
- **Component A (Maintenance Actions):** This typically includes:
 - **Checked/Tested:** Meters are tested for accuracy, often against a known standard. This can be done in a field spot-check or in a dedicated test bench.
 - **Recalibrated:** If a meter is found to be outside acceptable accuracy limits (e.g., $\pm 2-5\%$), it is adjusted to measure correctly.
 - **Replaced:** Meters that are faulty, beyond repair, or have reached the end of their economic life are swapped out for new ones.

KPI 17: Meters Checked, Recalibrated, and Replaced

- ❖ **Industry Standards for Frequency:** Utilities don't check 100% of meters yearly. Instead, they follow a planned program based on meter type, size, and age.
 - **Large Industrial/Commercial Meters:** Checked most frequently (e.g., every 1-2 years)
 - **Standard Residential Meters:** Follow a longer cycle (e.g., every 10 years), often based on the manufacturer's recommended service life.
- 1. **Audit Your Meter Population:** Classify meters by type, size, and installation date.
- 2. **Set a Maintenance Schedule:** Use industry standards as a guide (e.g., test large meters every 2 years, residential meters every 10 years).
- 3. **Calculate Your Annual Target:** If you have 10,000 residential meters on a 10-year cycle, your annual "A" target is roughly 1,000 meters.

Economic Efficiency & Financial Viability KPIs

Focuses on financial viability, cost recovery, and fiscal management.

Focus Area	Key Indicators	Goal
Cost Recovery	Operating Cost Coverage Ratio (OCCR); Working Ratio; Average Billed Tariff.	Ensure revenues exceed the cost of daily operations and maintenance.
Revenue Cycle	Revenue Collection Efficiency; Commercial Efficiency Index (CEI).	Minimize the gap between "water billed" and "cash in bank."
Cost Structure	Water Production Cost; Energy vs. O&M; Personnel Expenditure %.	Optimize input costs (power, chemicals, labor) to improve margins.
Asset & Debt	Maintenance Costs %; Debt Service Ratio (DSR).	Balance debt obligations while ensuring assets are not neglected.

WATER UTILITY KEY PERFORMANCE INDICATORS (KPIs)

TARIFF & REVENUE



OPERATING COST COVERAGE RATIO (OCCR)

Ability to cover operating expenses from operating revenues.



AVERAGE BILLED TARIFF

Average price billed per cubic meter of water.

$\$X.XX / m^3$



REVENUE COLLECTION EFFICIENCY

Percentage of billed revenue actually collected.



OPERATIONAL COSTS



WORKING RATIO (TARIFF COST RECOVERY)

Operating expenses relative to tariff revenue.



WATER PRODUCTION COST PER CUBIC METER

Cost to treat and distribute $1 m^3$ of water.

$\$Y.YY / m^3$



ENERGY COSTS AS A PERCENT OF TOTAL O & M COSTS



MAINTENANCE COSTS AS A PERCENT OF TOTAL O & M COSTS



EFFICIENCY & PERSONNEL

COMMERCIAL EFFICIENCY INDEX (CEI)



PERSONNEL EXPENDITURE AS PERCENTAGE OF REVENUE COLLECTION



FINANCIAL HEALTH



DEBT SERVICE RATIO (DSR)

Ability to make debt payments from operating income.

1.5x



KPI 18: Operating Cost Coverage Ratio (OCCR)

- Measures the financial viability of a utility by comparing recurring operational income against recurring operational expenditures.
- Determine if the utility can sustain its day-to-day services without relying on external capital injections or emergency government subsidies.

$$\text{OCCR} = (\text{Total Operating Revenue} \div \text{Total Operating Expenses (OPEX)}) * 100$$

- **Operating Revenue:** Includes all "cash-in" from core activities: water sales, sewerage charges, connection fees, and late penalties.
- **Note:** Non-operating income (like interest on investments or one-time asset sales) is typically excluded to show true operational strength.

KPI 18: Operating Cost Coverage Ratio (OCCR)

Operating Expenses: Includes all "cash-out" required for service delivery:

- Energy/Electricity for pumping.
- Chemicals for water treatment.
- Personnel (salaries and benefits).
- Routine maintenance and repairs.

Exclusion: Major capital projects (CAPEX) and debt principal repayments are usually excluded from this specific ratio.

Ratio Percentage	Sustainability Status	Strategic Action Required
> 120%	Strongly Sustainable	Surplus can be diverted to Capital Reserves or Debt Service.
100% – 110%	Marginal/Breakeven	Vulnerable to price shocks (e.g., a sudden rise in energy costs).
< 100%	Operating Deficit	Immediate need for tariff restructuring or operational cost-cutting.

KPI 19: Working Ratio (Tariff Cost Recovery)

- Measures the extent to which the annual revenue collected from user tariffs (bills) covers the total annual costs.
- A ratio of **1.0 or 100%** indicates that the tariff revenue is exactly sufficient to cover the operational costs and the capital investments.

$$\text{Working Ratio} = (\text{Total Billed Tariff Revenue Collected}) \div (\text{Total O\&M Cost} + \text{Total E\&I Cost})$$

- **Expansion & Improvement (E&I) Costs:** This is the **capital expenditure (CapEx)** incurred in that specific year. It includes:
 - **Expansion:** New pipelines, plants, or connections to serve more customers.
 - **Improvement:** Upgrades, replacements, and major rehabilitation of existing assets.

KPI 19: Working Ratio (Tariff Cost Recovery)

Ratio > 1.0 (or >100%):

- Tariff revenue exceeds annual costs.
- Generating a **cash surplus**.

Ratio = 1.0 (or 100%):

- Perfect cost recovery. Revenue exactly matches costs.
- **Self-sustaining** for that year. It meets its financial obligations without surplus or deficit.

Ratio < 1.0 (or <100%):

- Tariff revenue is insufficient to cover annual costs.
- Running a **cash deficit**. Leads to deteriorating service, deferred maintenance, and an inability to expand.

KPI 20: Average Billed Tariff (Birr/m³)

- Average price charged to customers per unit of water (m³) for which a bill was issued.
- Focuses strictly on **invoiced** amounts and **metered** sales:

$$\text{Average Billed Tariff} = \text{Total Water Sales Billed} \div \text{Total Volume of Water Sold}$$

- **Numerator:** Total amount of all water bills sent to customers in the period.
- **Denominator:** The total volume of water recorded on all customer meters.
- **Cost Recovery Analysis:** tells you if the price you are charging is enough to cover your Unit Cost of Production.

KPI 20: Average Billed Tariff (Birr/m³)

- **Tariff Structure Effectiveness:** Since most utilities use Increasing Block Tariffs, this KPI shows the "effective rate."
- **Revenue Forecasting:** If you plan to increase water supply by 10%, you can multiply that volume by the Average Billed Tariff to predict new revenue.
- **The "O&M" Benchmark:** A healthy utility should have an ideal Average Billed Tariff that is greater than 120% of the O&M costs.
- **Social Benchmark:** To remain affordable, the bill for a "Basic Needs" volume (5m³ per month) should not exceed **3%-5%** of the average household income.

KPI 21: Revenue Collection Efficiency

- Measures a utility's effectiveness in converting billed receivables into liquid capital.

$$\text{Collection Efficiency (\%)} = (\text{Total Revenue Collected} \div \text{Total Amount Billed}) * 100$$

- **Total Revenue Collected:** Sum of cash, electronic transfers, and mobile payments received and reconciled during the reporting period. Includes the recovery of arrears.
- **Total Amount Billed:** Gross value of all water and sanitation invoices generated during the same period, including fixed service charges and volumetric fees.

KPI 21: Revenue Collection Efficiency

Efficiency (%)	Performance Status	Strategic Meaning
> 98%	Optimal	Indicates high customer satisfaction and seamless payment gateways.
90% – 97%	Acceptable	Standard for most urban utilities; suggests minor friction in collection cycles.
80% – 89%	Underperforming	Potential issues with billing accuracy or ineffective debt recovery policies.
< 80%	Critical	High risk of insolvency; suggests systemic data errors or widespread customer non-compliance.

- High collection efficiency ensures the utility can meet immediate obligations, such as energy and chemical procurement.
- A declining ratio serves as an early warning sign to tighten measures, such as late payment penalties or service disconnections.
- Billing disputes (non-payment) often indicate a lack of confidence in meter accuracy or service quality.

KPI 22: Commercial Efficiency Index (CEI)

- The total volume of water produced that is actually **paid for** by the customers.
- Looks at the finish line: **cash in the bank**. Accounts for both NRW and unpaid bills.
- Calculated by multiplying the **Billing Efficiency** by the **Collection Efficiency**:

$$\text{CEI} = ((\text{Volume Billed} \div \text{Volume Produced}) * (\text{Amount Collected} \div \text{Amount Billed})) * 100$$

- **Billing Efficiency:** How much of the water produced is actually billed.
- **Collection Efficiency:** How much of the money billed to customers is actually collected.

KPI 22: Commercial Efficiency Index (CEI)

Performance Level	CEI Range	Status
Excellent	> 95%	Highly sustainable; minimal non-revenue water and high recovery.
Good	85% – 95%	Standard for well-managed utilities in developing regions.
Fair	70% – 85%	Significant room for improvement in meter reading or debt recovery.
Poor	< 70%	High risk of financial insolvency; likely heavy subsidies required.

Management Strategies to Improve CEI:

- **Meter Replacement Programs:** Replacing aged meters that "under-read" consumption.
- **GIS Mapping:** Ensuring every connection has a corresponding account in the billing database.
- **Incentivized Collection:** Offering easy payment options (mobile) to increase the collection.

KPI 23: Water Production Cost per Cubic Meter

- Calculated by dividing the utility's total annual running expenses by the total volume of water sold.

$$\text{Total Annual Production Cost (Birr)} \div \text{Total Annual Volume of Water Sold (m}^3\text{)} = \text{Cost per Cubic Meter (Birr/m}^3\text{)}$$

The "total production costs" typically include:

- **Personnel Costs:** Salaries for operators, technicians, and administrative staff.
- **Energy Costs:** The single largest operational expense, primarily for running pumps.
- **Chemical Costs:** For water treatment (e.g., coagulants, chlorine for disinfection).
- **Maintenance & Parts:** Regular servicing, repairs, and replacement of components.
- **Other Consumables & Utilities:** Costs for water used in processes, communications, and waste handling.

KPI 23: Water Production Cost per Cubic Meter

- **Tariff Setting:** Production cost exceeds your billing rate, the utility is bleeding money. Identifies the need for subsidies or tariff setting.
- **Operational Efficiency:** Sudden spikes usually signal underlying issues, such as equipment failure, poor chemical dosing, or rising energy prices.
- **Budgeting:** It allows for precise forecasting. If you know your demand will grow by 5%, you can use this KPI to estimate the exact increase in variable costs.
- **High Performers:** Typically maintain a high "OCCR," ensuring that the cost per cubic meter is significantly lower than the revenue collected.
- **To Improve this KPI:** Energy Audits, NRW Reduction, Automation etc.

KPI 24: Personnel Expenditure as Percentage of Revenue Collection

- What portion of the money collected from customers is spent on staff salaries and benefits?

$$\text{Personnel Expenditure as \% of Revenue Collection} = (\text{Total Personnel Costs} \div \text{Total Collected Operating Revenue}) * 100$$

- Primary purpose is to assess the balance between a utility's largest operational cost (its workforce) and its core income stream.
- A lower percentage indicates greater efficiency and more revenue available for infrastructure investment, debt service, and other operational costs.

KPI 24: Personnel Expenditure as Percentage of Revenue Collection

Component	What to Include	Data Source & Notes
Numerator: Personnel Expenditure	Salaries, wages, bonuses, pensions, health insurance, taxes, and all other employee benefits	Financial accounts (Profit & Loss Statement). Ensure it covers all departments
Denominator: Revenue Collection	All cash actually collected from: water bills, connection fees, late payment charges, and other operational service fees.	Crucially, this is cash-based, not billed revenue. It should reflect collections in the same period as the personnel costs.

- **Typical Ranges:** For efficient, mature utilities in developed nations, this figure often falls between **20% and 35%**.
- **High Percentage (>40-50%):** This can signal overstaffing, inefficient processes, or critically **low revenue collection rates**.
- **Very Low Percentage (<15%):** May indicate **under-investment in skilled staff**, excessive outsourcing, or could be a feature of a very large, automated utility.

KPI 25: Energy Costs as a Percent of O&M Cost

$$\text{Energy vs. O\&M Cost \%} = (A / B) \times 100$$

- **A** = Total annual energy cost (Birr)
 - **B** = Total utility's annual costs/expenditures (Birr)
-
- 20% - 40%: The world Bank identifies this is a common global range for the energy share of operating costs
 - A higher percentage indicates a utility is highly energy-intensive. This makes its finances very sensitive to electricity price hikes.
 - A lower percentage may suggest a gravity-fed system or that other costs (like labor or chemicals) are comparatively higher.

KPI 25: Energy Costs as a Percent of O&M Cost

- **Sensitivity to Tariffs:** Electricity is often a "price-taker" cost. If ratio is high, a minor climb in national electricity tariffs could instantly push the utility into a deficit.
- **Operational Efficiency Indicator:** High energy costs (above 40%) relative to O&M often signal aging infrastructure, such as inefficient pumps or high NRW. Urgent energy audit required.
- **Recommendations for Improvement:** Conduct pump efficiency tests, NRW reduction, solar integration

KPI 26: Maintenance Costs as a Percent of Total Operating Costs

- Measures the financial weight of maintaining assets (pumps, pipes, treatment plants) relative to the entire cost of running the utility.

$$\text{Maintenance \%} = (\text{Total Annual Maintenance Costs} \div \text{Total Annual Operating Costs}) * 100$$

- **High percentage:** Might indicate aging infrastructure requiring frequent "firefighting" repairs or a robust proactive maintenance strategy.
- **Low percentage:** Could suggest high operational efficiency, or conversely, a dangerous neglect of assets that might lead to catastrophic failure later.

KPI 26: Maintenance Costs as a Percent of Total Operating Costs

Component	What it includes
Maintenance Costs	Spare parts, maintenance staff wages, external contractor fees, lubricants, and diagnostic tools.
Total Operating Costs (OPEX)	Maintenance costs + energy/electricity + chemicals + administrative salaries + insurance + taxes.

- **Budgeting:** It helps managers decide if they need to shift funds from administration to the field.
- **Asset Health:** If this percentage rises year-over-year without an increase in service, your infrastructure is likely reaching the end of its life cycle.
- **Benchmarking:** Compare your utility's performance against industry standards (typically ranging between **15% to 25%** depending on the region and age of the system).

KPI 27: Debt Service Ratio (DSR)

- Measures the utility's ability to cover its debt obligations (interest and principal) using its total generated revenue.

$$\text{DSR} = (A \div B) * 100$$

- **A = Total Annual Debt Service (Birr):** The sum of all principal and interest payments due within the fiscal year.
- **B = Total Revenue in a Year (Birr):** The total income collected, primarily from water tariffs, connection fees, and other service charges.
- **Ideal Range:** Most utilities aim for a ratio where revenue is significantly higher than debt service. The ideal range is DSR less than **20% to 25%**.

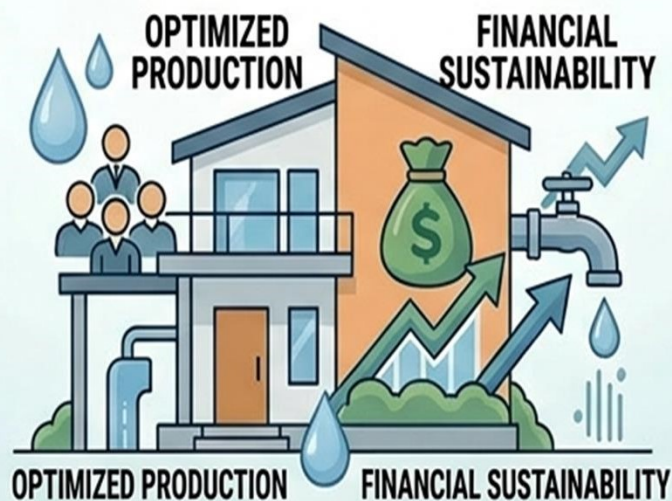
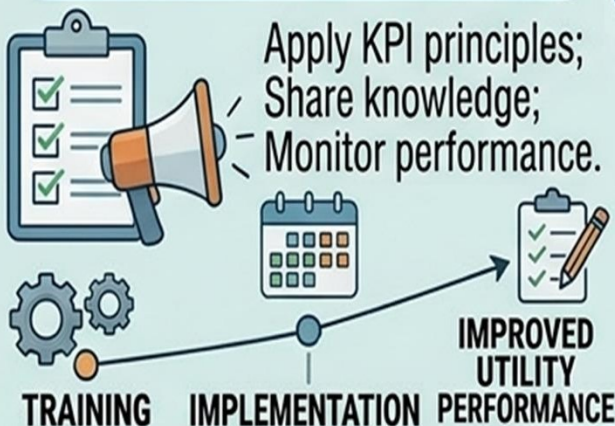
WATER UTILITY KPI TRAINING: FINAL SUMMARY & CONCLUSION

KEY KPI TOPICS REVIEWED



Key performance indicators for financial and operational success.

NEXT STEPS & ACTION PLAN



THANK YOU FOR ADVANCING WATER UTILITY PERFORMANCE. Finalized Presentation.

TRAINING OUTCOMES & IMPACT



ENHANCED STRATEGIC

Ability to use KPIs collection data for decision-making.

ENHANCED STRATEGIC PLANNING

Ability to use KPIs for long-term utility goals.

OPTIMIZED RESOURCE ALLOCATION

Knowledge to assign budget and personnel based on performance data.

IMPROVED DECISION-MAKING

Confidence to make data-driven decisions.

THANK YOU & RECOGNITION

THANK YOU, TRAINEES!

For your engagement, participation, and commitment to data excellence. Your dedication to your utility is appreciated.



WATER UTILITY KPI MASTERY RECOGNIZED

YIRGA ALEMU